Section 9	Bill of Quantities

I N D E X

SL. NO	Description	Page No
1.	BOQ	269-282

NAME OF WORK : Baiyyappanahalli - Hosur Doubling Project - "Proposed Construction of Staff Quartets Type-II - 8 units and Type-III - 8 units at Hosuru Railway Station and other miscellaneous works including Electrification works in Bangalore Division of South Western Railway.

SL. NO	SCHEDULE	DESCRIPTION OF SCHEDULE	AMOUNT
1	Sch "A"	Item of work covered under USSOR of 2011 SWR	122107
2	Sch "A1"	Item of work covered under Schedule of Rates 2018 of CPWD , Delhi - The rate is inclusive of Supply of Cement	19572961
3	Sch "E"	Items of works pertains to Supply of Steel	2330550
4	Sch "J"	Miscellaneous Items	2063855
5	Sch "E1"	Internal Electrification of Staff Quarters at Hosur Railway Station	1442252
6	Sch "E2"	Lighting arrangements for approach road for Quarters at Hosur Railway station	295761
TOTAL			**25827485**

Seal and Signature of the Tenderer

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NAME OF WORK : Baiyyappanahalli - Hosur Doubling Project - "Proposed Construction of Staff Quartets Type-II - 8 units and Type-III - 8 units at Hosuru Railway Station and other miscellaneous works including Electrification works in Bangalore Division of South Western Railway.

Schedule - "A"						
SL. NO	**Item Code**	**Description of Items**	**Unit**	**Rate**	**Qty**	**Amount**
1	162129	Avenue Plants : Any other kind of tree of height 210-240-cm in gunny bag	Each	88.27	10.00	882.70
2	161020	Supplying and stacking of good earth at site including royalty (earth measured in stacks will be reduced by 20% for payment)	cum	180.13	50.00	9006.50
3	161041	Supplying and stacking at site dump manure / farm manure / animal dung manure from approved source; including all lead and lift etc. (manure measured in stacks will be reduced by 8% for payment) : Screened through sieve of I.S. designation 20mm	cum	106.81	10.00	1068.10
4	161210	Excavation in dumped stones or malba/ garbage / rubbish including stacking of serviceable and unserviceable material separately and disposal of unserviceable material lead upto 50metre and lift upto 1.5metre and disposed material to be neatly dressed.	cum	166.68	10.00	1666.80
5	191030	Extra for 150mm dia bore in hard Rock/ large Boulder at all levels	No	4878.30	20.00	97566.00
6	191100	Taking out 100mm dia & 450mm long undisturbed samples of soil from bore holes, including provision of air tight containers for packing and, labelling incl. transporting the samples to laboratory. Piston sampler shall be used for extracting undisturbed sam	No	348.45	2.00	696.90
7	191120	Conducting standard penetration test as per IS 2131 at approximate1.5 m intervals in bore holes, as directed by the Engineer in charge	NO	209.07	2.00	418.14
8	191141	Moisture contents/ Dry density	Each	290.38	2.00	580.76
9	191142	Conducting labouratary tests on collected soil samples Atterberg limit	Each	290.38	2.00	580.76
10	191143	Conducting labouratary tests on collected soil samples: specific gravity	Each	232.30	2.00	464.60
11	191144	Conducting labouratary tests on collected soil samples: Grain size analysis	Each	232.30	2.00	464.60
12	191145	Conducting labouratary tests on collected soil samples: Direct shear test	Each	1742.25	2.00	3484.50
13	191146	Conducting labouratary tests on collected soil samples: Natural density	Each	290.38	2.00	580.76

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Schedule - "A"						
SL. NO	**Item Code**	**Description of Items**	**Unit**	**Rate**	**Qty**	**Amount**
14	191147	Conducting labouratary tests on collected soil samples: consolidation test	Each	290.38	2.00	580.76
15	191148	Conducting labouratary tests on collected soil samples): Unconfined compression test	Each	290.38	2.00	580.76
16	191149	Conducting labouratary tests on collected soil samples): Tri axial test	Each	1742.25	2.00	3484.50
No. of Items Sixteen only		**SCHEDULE "A" Departmental Value**				**122107**
		Tenderer's / Contractor's percentage				
		SCHEDULE "A" Total				

Seal and Signature of the Tenderer

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NAME OF WORK : Baiyyappanahalli - Hosur Doubling Project - "Proposed Construction of Staff Quartets Type-II - 8 units and Type-III - 8 units at Hosuru Railway Station and other miscellaneous works including Electrification works in Bangalore Division of South Western Railway.

Schedule - "A1"						
SL. NO	**Item Code**	**Description of Items**	**Unit**	**Rate**	**Qty**	**Amount**
1	2.8.1	Earth work in excavation by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift upto 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 m : All kinds of soil.	cum	252.3	550	138765.00
2	2.9.1	Excavation work by mechanical means (Hydraulic excavator)/ manual means in foundation trenches or drains (not exceeding 1.5m in width or 10 sqm on plan), including dressing of sides and ramming of bottoms, lift upto 1.5 m, including getting out the excavated soil and disposal of surplus excavated soils as directed, within a lead of 50 m : Ordinary rock	cum	448.15	50	22407.50
3	2.26	Extra for every additional lift of 1.5 m or part thereof in excavation / banking excavated or stacked materials : All kinds of soil	cum	90.4	400	36160.00
4	2.27	Supplying and filling in plinth with sand under floors, including watering, ramming, consolidating and dressing complete.	cum	1953.05	100	195305.00
5	2.31	Clearing jungle including uprooting of rank vegetation, grass, brush wood, trees and saplings of girth up to 30 cm measured at a height of 1 m above ground level and removal of rubbish up to a distance of 50 m outside the periphery of the area cleared.	sqm	12.55	100	1255.00
6	2.36	Extra for levelling & neatly dressing of disposed soil completely as directed by Engineer-in-charge.	cum	66.35	50	3317.50
7	4.1.6	Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work up to plinth level : 1:3:6 (1 Cement : 3 coarse sand (zone-III): 6 graded stone aggregate 40 mm nominal size)	cum	6126.25	60	367575.00
8	4.7.1	P/L precast cement concrete solid blocks: 1:1.5:3 (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20 mm nominal size)	cum	14241.05	350	4984367.50
9	5.1.2	Providing and laying in position specified grade of reinforced cement concrete, excluding the cost of centering, shuttering, finishing and reinforcement - All work up to plinth level : 1:1.5:3 (1 cement : 1.5 coarse sand (zone-III): 3 graded stone aggregate 20 mm nominal size)	cum	7718.25	85	656051.25
10	5.2.2	Reinforced cement concrete work in walls (any thickness), including attached pilasters, buttresses, plinth and string courses, fillets, columns, pillars, piers, abutments, posts and struts etc. above plinth level up to floor five level, excluding cost of centering, shuttering, finishing and reinforcement : 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) : 3 graded stone aggregate 20 mm nominal size)	cum	9306	400	3722400.00
11	5.9.1	Centering and shuttering including strutting, propping etc. and removal of form for :Foundations, footings, bases of columns, etc. for mass concrete	sqm	284.85	600	170910.00
12	5.9.3	Centering and shuttering including strutting, propping etc. and removal of form for : Suspended floors, roofs, landings, balconies and access platform	sqm	693.05	1600	1108880.00
13	5.9.6	Columns, Pillars, Piers, Abutments, Posts and Struts	sqm	733.7	600	440220.00

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Schedule - "A1"						
SL. NO	**Item Code**	**Description of Items**	**Unit**	**Rate**	**Qty**	**Amount**
14	5.11.1	Extra for additional height in centering, shuttering where ever required with adequate bracing, propping etc., including cost of de-shuttering and decentering at all levels, over a height of 3.5 m, for every additional height of 1 metre or part thereof (Plan area to be measured) : Suspended floors, roofs, landing, beams and balconies (Plan area to be measured)	sqm	287.5	1000	287500.00
15	5.23	Smooth finishing of the exposed surface of R.C.C. work with 6 mm thick cement mortar 1:3 (1 Cement : 3 fine sand)	sqm	227.35	1900	431965.00
16	5.30	Add for plaster drip course/ groove in plastered surface or moulding to R.C.C. projections.	metre	58.9	300	17670.00
17	5.34.1	Extra for providing richer mixes at all floor levels. Note:- Excess/less cement over the specified cement content used is payable /recoverable separately. : Providing M-30 grade concrete instead of M-25 grade BMC/ RMC. (Note:- Cement content considered in M-30 is @ 340 kg/cum)	cum	69.75	200	13950.00
18	5.35	Add for using extra cement in the items of design mix over and above the specified cement content therein.	quintal	673.3	75	50497.50
19	9.1	Providing wood work in frames of doors . Windows , clerstory windows and other frames , wrouht framed fixied in position with old fast lugs or with dash fsteners required dia and & length (hold fast lugs or dash fastener shall be paid for saparatly) 91.1.) Second class teak wood	Cum	130183.05	0.75	97637.29
20	9.16.11	Providing and fixing 25 mm thick glazed shutters Second class teak wood including ISI marked anodised aluminium butt hinges with necessary screws	sqm	3794.85	100	379485.00
21	9.12	Providing and fixing factory made panel PVC door shutter consisting offrame made out of M.S. tubes of 19 gauge thickness and size of 19 mm x 19 mm for styles and 15x15 mm for top & bottom rails. M.S. frame shall have a coat of steel primers of approved make and manufacture. M.S. frame covered with 5 mm thick heat moulded PVC 'C' channel of size 30 mm thickness, 70 mm width out of which 50 mm shall be flat and 20 mm shall be tapered in 45 degree angle on both side forming styles and 5 mm thick, 95 mm wide PVC sheet out of which 75mm shall be flat and 20 mm shall be tapered in 45 degree on the inner side to form top and bottom rail and 115 mm wide PVC sheet out of which 75 mm shall be flat and 20 mm shall be tapered on both sides to form lock rail. Top, bottom and lock rails shall be provided both side of the panel. 10 mm (5 mm x 2) thick, 20 mm wide cross PVC sheet be provided as gap insert for top rail & bottom rail, paneling of 5 mm thick both side PVC sheet to be fitted in the M.S. frame welded/ sealed to the styles & rails with 7 mm (5 mm+2 mm) thick x 15 mm wide PVC sheet beading on inner side, and joined together with solvent cement adhesive. An additional 5 mm thick PVC strip of 20 mm width is to be stuck on the interior side of the 'C' Channel using PVC solvent adhesive etc. complete as per direction of Engineer-in-charge, manufacturer's specification & drawing : 30 mm thick pre laminated PVC door shutters	sqm	2423.4	25	60585.00
22	9.48.1	Providing and fixing M.S. grills of required pattern in frames of windows etc. with M.S. flats, square or round bars etc. including priming coat with approved steel primer all complete : Fixed to steel windows by welding	kg	165.3	1100	181830.00

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Schedule - "A1"						
SL. NO	**Item Code**	**Description of Items**	**Unit**	**Rate**	**Qty**	**Amount**
23	9.119	Providing and fixing factory made P.V.C. door frame of size 50x47 mm with awall thickness of 5 mm, made out of extruded 5mm rigid PVC foam sheet, mitred at corners and joined with 2 Nos of 150 mm long brackets of 15x15 mm M.S. square tube, the vertical door frame profiles to be reinforced with 19x19 mm M.S. square tube of 19 gauge, EPDM rubber gasket weather seal to be provided through out the frame. The door frame to be fixed to the wall using M.S. screws of 65/100 mm size, complete as per manufacturer's specification and direction of Engineerin-Charge.	metre	391.65	70	27415.50
24	9.147D.2	Providing and fixing factory made uPVC white colour sliding glazed window upto 1.50 m in height dimension comprising of uPVC multi-chambered frame with in-built roller track and sash extruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made from roll forming process of required length (shape & size according to uPVC profile), appropriate dimension of uPVC extruded glazing beads and uPVC extruded interlocks, EPDM gasket, wool pile, zinc alloy (white powder coated) touch locks with hook, zinc alloy body with single nylon rollers (weight bearing capacity to be 40 kg), G.I fasteners 100 x 8 mm size for fixing frame to finished wall and necessary stainless steel screws etc. Profile of frame & sash shall be mitred cut and fusion welded at all corners, including drilling of holes for fixing hardware's and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealent over backer rod of required size and of approved quality, all complete as per approved drawing & direction of Engineer-in-Charge. (Single / double glass panes, wire mesh and silicon sealent shall be paid separately): Three track three panels sliding window with fly proof SS wire mesh (Two nos. glazed & one no. wire mesh panels) made of (small series) frame 92 x 44 mm & sash 32 x 60 mm both having wall thickness of 1.9 ± 0.2 mm and single glazing bead of appropriate dimension (Area of window upto 1.75 sqm).	sqm	8954.2	90	805878.00
25	10.28	Providing and fixing stainless steel (Grade 304) railing made of Hollow tubes, channels, plates etc., including welding, grinding, buffing, polishing and making curvature (wherever required) and fitting the same with necessary stainless steel nuts and bolts complete, i/c fixing the railing with necessary accessories & stainless steel dash fasteners , stainless steel bolts etc., of required size, on the top of the floor or the side of waist slab with suitable arrangement as per approval of Engineer-incharge, (for payment purpose only weight of stainless steel members shall be considered excluding fixing accessories such as nuts, bolts, fasteners etc.)	kg	575.45	50	28772.50
26	11.7	Cement concrete pavement with 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size), including finishing complete	cum	7335.1	50	366755.00
27	11.37	Providing and laying Ceramic glazed floor tiles of size 300x300 mm (thickness to be specified by the manufacturer) of 1st quality conforming to IS : 15622 of approved make in colours such as White, Ivory, Grey, Fume Red Brown, laid on 20 mm thick cement mortar 1:4 (1 Cement : 4 Coarse sand), Jointing with grey cement slurry @ 3.3 kg/sqm including pointing the joints with white cement and matching pigment etc., complete	sqm	926.9	200	185380.00

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Schedule - "A1"						
SL. NO	**Item Code**	**Description of Items**	**Unit**	**Rate**	**Qty**	**Amount**
28	11.49.2	Providing and laying Vitrified tiles in floor with different sizes (thickness to be specified by the manufacturer), with water absorption less than 0.08% and conforming to IS:15622, of approved brand & manufacturer, in all colours and shade, laid with cement based high polymer modified quick set tile adhesive (water based) conforming to IS : 15477, in average 6 mm thickness, including grouting of joints (Payment for grouting of joints to be made separately) : Size of Tile 600x600 mm	sqm	1609.95	450	724477.50
29	11.50	Deduct for not grouting the joints with white cement and matching pigment in the items of fixing of vitrified tiles	sqm	11	850	9350.00
30	11.55.1	Providing and laying flamed finish Granite stone flooring in required design and patterns, in linear as well as curvilinear portions of the building all complete as per the architectural drawings with 18 mm thick stone slab over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) laid and jointed with cement slurry and pointing with white cement slurry admixed with pigment of matching shade including rubbing, curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge : Flamed finish granite stone slab Jet Black, Cherry Red, Elite Brown, Cat Eye or equivalent.	sqm	2476.45	20	49529.00
31	12.41.1	Providing and fixing on wall face unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion, (i) Single socketed pipes : 75 mm diameter	metre	201.1	200	40220.00
32	12.41.2	Providing and fixing on wall face unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion, (i) Single socketed pipes : 110 mm diameter	metre	305.05	300	91515.00
33	13.1.1	12 mm cement plaster of mix : 1:4 (1 cement: 4 fine sand)	sqm	266.85	1500	400275.00
34	13.2.1	15 mm cement plaster on the rough side of single or half brick wall of mix : 1:4 (1 cement: 4 fine sand)	sqm	307.9	1000	307900.00
35	13.16.1	6 mm cement plaster of mix : 1:3 (1 cement : 3 fine sand)	sqm	227.35	6000	1364100.00
36	13.21	Extra for providing and mixing water proofing material in cement plaster work in proportion recommended by the manufacturers	per bag of 50kg cement used in the mix	59.75	140	8365.00
37	13.22	Extra for plastering exterior walls of height more than 10 m from ground level for every additional height of 3 m or part thereof.	sqm	65.95	800	52760.00
38	13.28.3	12 mm thick plain cement mortar bands in cement mortar 1:4 (1 cement : 4 fine sand): Raised Band	cm per metre	7.1	1700	12070.00
39	13.45	Finishing walls with textured exterior paint of required shade : New work (Two or more coats applied @ 3.28 ltr/10 sqm) over and including priming coat of exterior primer applied @ 2.20kg/10 sqm	sqm	232	750	174000.00

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